Support Vector Machine

1. Why does maximizing margin improve generalization? (VC Dimension View)

For a linear classifier:

$$f(x) = w^T x + b$$

The **geometric margin** is:

$$\gamma = \frac{1}{\| w \|}$$

Key Theory Result

For data bounded in radius R:

$$VC\text{-dim} \leq \min\left(\frac{R^2}{\gamma^2}, d\right) + 1$$

So:

- Larger margin (↑ γ)
- Smaller $\| w \|$
- Lower VC dimension
- Lower model capacity
- Better generalization bounds

Intuition

Maximizing margin:

- Makes classifier robust to perturbations
- Reduces sensitivity to noise
- Controls hypothesis complexity directly

This connects directly to **capacity control**, not just empirical fit.

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2. Dual Form of SVM (Using Lagrange Multipliers)

Primal (Hard Margin)

$$\min_{w,b} \frac{1}{2} \| w \|^2$$

Subject to:

$$y_i(w^T x_i + b) \geq 1$$

Lagrangian

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \| w \|^2 - \sum_i \alpha_i \left[y_i(w^T x_i + b) - 1 \right]$$

with $\alpha_i \geq 0$

Stationarity Conditions

$$w = \sum_i \alpha_i \, y_i x_i$$

$$\sum_i \alpha_i \, y_i = 0$$

Dual Problem

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \, \alpha_j y_i y_j x_i^T x_j$$

Subject to:

$$\alpha_i \geq 0, \sum_i \alpha_i \, y_i = 0$$

Important Insight

The solution depends only on **dot products** $x_i^T x_j$ → enables **kernel trick**.

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3. Why Only Support Vectors Matter?

From KKT conditions:

$$\alpha_i[y_i(w^T x_i + b) - 1] = 0$$

So:

- If point is outside margin → constraint inactive → $\alpha_i = 0$
- If on margin → $\alpha_i > 0$

Final classifier:

$$f(x) = \sum_{i \in SV} \alpha_i \, y_i K(x_i, x) + b$$

Only support vectors have non-zero multipliers.

Interpretation

SVM is sparse.
Decision boundary determined only by boundary points.

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4. Hinge Loss vs Logistic Loss

Hinge Loss (SVM)

$$\max\,(0, 1 - yf(x))$$

Properties:

- Convex
- Not differentiable at 1
- Encourages margin maximization

- Produces sparse solutions

Logistic Loss (Logistic Regression)

$$\log\left(1 + e^{-yf(x)}\right)$$

Properties:

- Smooth
- Probabilistic interpretation
- Penalizes all points (no sparsity)

Geometric Difference

- Hinge → linear penalty beyond margin
- Logistic → exponential tail

Hinge focuses on margin violations only.
Logistic pushes all points.

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5. Convexity of SVM Objective

Soft-margin SVM:

$$\min_{w,b} \frac{1}{2} \| w \|^2 + C \sum_i \max\left(0, 1 - y_i f(x_i)\right)$$

- $\| w \|^2$ is convex (quadratic form)
- Hinge loss is convex
- Sum of convex functions is convex

Therefore:

- Global optimum guaranteed
- No local minima

- Strong duality holds

This is a major advantage over neural networks.

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6. Kernel Choice & Feature Space Geometry

Kernel:

$$K(x, z) = \phi(x)^T \phi(z)$$

Kernel defines:

- Geometry of feature space
- Inner product structure
- Implicit embedding

Examples:

Linear Kernel

No feature expansion.

Polynomial Kernel

$$K(x, z) = (x^T z + c)^d$$

Creates curved boundaries.

RBF Kernel

$$K(x, z) = e^{-\gamma \|x - z\|^2}$$

Maps to infinite-dimensional space.

Geometric Insight

Kernel defines similarity metric.
RBF = local neighborhoods.
Polynomial = global interactions.

Choice affects:

- Smoothness
- Decision boundary curvature
- Effective VC dimension

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7. What Happens When $C \to \infty$?

Soft-margin objective:

$$\frac{1}{2} \| w \|^2 + C \sum \xi_i$$

As $C \to \infty$:

- Misclassification heavily penalized
- Model tries to fit training data perfectly
- Approaches hard-margin SVM (if separable)
- Risk of overfitting increases

Large C:
Low bias, high variance.

Small C:
High bias, low variance.

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8. SVM from Structural Risk Minimization (SRM)

From **Vladimir Vapnik**'s SRM principle:

Goal:
Minimize

$$R(f) \leq R_{emp}(f) + \text{capacity term}$$

Capacity measured by VC dimension.

For linear classifiers:

$$h \leq \frac{R^2}{\gamma^2}$$

Thus minimizing:

- Empirical hinge loss
- While maximizing margin (γ)

Is equivalent to minimizing upper bound on true risk.

SVM is a direct implementation of SRM.

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9. SVM vs Neural Networks in High Dimensions

In High-Dimensional Regimes (p >> n)

SVM:

- Strong regularization via margin
- Convex optimization
- Works well with kernels
- Effective with small datasets

Neural Networks:

- Non-convex optimization
- Implicit regularization via SGD
- Require larger data
- Scale better with very large data

Modern Insight

In extremely high dimensions:

- Linear SVM often competitive
- Especially in text classification
- Sparse data favors SVM

Deep networks outperform when:

- Representation learning is needed
- Data is abundant

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10. When Does SVM Fail in Practice?

1. Very large datasets ($O(n^2)$ memory in kernel SVM)
2. Highly overlapping classes
3. Noisy labels with large C
4. Poor kernel choice
5. Imbalanced datasets (needs weighting)
6. When probabilistic outputs required (needs calibration)
7. Very high noise near decision boundary

Final Conceptual Summary

SVM is:

- A margin maximizer
- A convex optimization problem
- A sparse solution method
- A kernel-based geometric classifier
- A practical implementation of SRM theory

It bridges:

- Geometry
- Optimization
- Statistical learning theory